Is Handwriting Still Human? A Unified Benchmark of AI and Procedural Handwriting Synthesis

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***Abstract*—Despite significant advances in handwriting synthesis, no study has yet systematically compared the perceived realism of current techniques under identical photorealistic rendering conditions. This paper presents the comparative evaluation of text-to-handwriting conversion across four categories: (i) non-learning procedural methods, (ii) popular commercial platforms (2025 free tiers), (iii) state-of-the-art open-source deep models retrained on an extended multilingual corpus, and (iv) real human handwriting as ground truth.**

We retrain and benchmark representative models from three major paradigms — recurrent (LSTM+MDN), Transformer-based (Handwriting Transformer), and diffusion-based (DiffuHand 2025) — on a merged dataset combining IAM-OnDB, CVL, and RIMES (English, German, French). All outputs are rendered with a unified photorealistic pipeline (paper texture, ink bleed, micro-defects).

Key contributions: 1) A unified evaluation framework and photorealistic post-processing pipeline applied fairly to all categories; 2) The largest blind Turing test to date on handwriting realism (249 participants , conducted March–April 2025); 3) Quantitative and perceptual comparison showing that modern open-source diffusion models now outperform both commercial free tools and traditional procedural methods, reaching human fooling rates above 82 4) Updated multilingual benchmarks and ready-to-use weights on IAM+CVL+RIMES.

Results establish a new state-of-the-art baseline for accessible, high-realism handwriting synthesis in 2025.

I. Introduction

Handwritten text carries unique emotional and aesthetic value that digital typography cannot replicate [1]. Automatic synthesis of realistic handwriting from digital text therefore has applications in education, accessibility, digital humanities, forensic counter-measures, and personalized communication.

The field has progressed from rule-based fonts and Bézier-curve engines to deep generative models capable of remarkable visual fidelity [2], [12]. However, most evaluations remain fragmented: procedural methods, commercial tools, and academic models are rarely compared directly, and almost never under realistic rendering conditions (paper texture, ink behaviour, scanning artefacts).

This work addresses these limitations through a systematic, large-scale empirical study conducted in 2025. We compare four categories of handwriting synthesis:

- Non-learning procedural baselines (enhanced handwriting fonts + SVG stroke engines)
- Commercial web platforms (free tiers of Calligrapher.ai, 10015.io, HandtextAI, etc.)
- State-of-the-art open-source models retrained on a merged multilingual corpus (IAM-OnDB + CVL + RIMES)
- Real human handwriting (ground truth)

All generated samples undergo identical photorealistic post-processing. Perceptual realism is assessed via a blind Turing test with 249 participants .

The main contributions are:

1) A reproducible evaluation framework and photorealistic rendering pipeline;
2) New multilingual benchmarks for three major open-source paradigms;
3) The largest human evaluation of handwriting realism to date;
4) Practical insights on current detectability limits and ethical considerations.

II. Related Work

The synthesis of realistic handwriting from digital text has evolved through several major technological paradigms.

Early approaches (1990–2015) relied on static font concatenation and rule-based ligatures, yielding visually acceptable but mechanically rigid results devoid of natural variation in pressure, speed, or inter-character connections [1].

The introduction of sequence modelling brought a significant leap forward. Recurrent neural networks, particularly Long Short-Term Memory (LSTM) models with mixture density network outputs [2], enabled the first dynamic stroke-level generation using the IAM-OnDB dataset [7]. Subsequent improvements incorporated attention mechanisms [3] and variational modelling to enhance long-range consistency.

Generative Adversarial Networks (GANs) markedly improved perceptual quality. Notable examples include GAN-writing [10], DeepWriting [3], and style-transfer frameworks , some of which reported human detection rates below 20 % in controlled settings.

More recently, Transformer-based architectures [12] and especially diffusion models have pushed raster-level photoreal-

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ism and few-shot personalisation to new levels, often requiring only a handful of reference lines.

Despite substantial progress, several important gaps remain in the literature:

- Most perceptual evaluations are typically conducted on small cohorts or with synthetic metrics (FID, stroke error) rather than large-scale human studies under realistic viewing conditions;
- photorealistic rendering aspects (paper texture, ink bleed, lighting, micro-defects) are rarely standardised or jointly evaluated;
- direct, fair comparisons between non-learning procedural methods, commercial platforms, open-source academic models, and real human handwriting are virtually absent, especially on multilingual data;
- ethical implications and current forensic detection limits receive limited systematic analysis.

The present work helps close these gaps by:

1) proposing a unified evaluation framework with consistent photorealistic post-processing applied to all methods;
2) retraining and benchmarking three major open-source paradigms (LSTM+MDN, Transformer-based, and diffusion-based) on an extended multilingual corpus combining IAM-OnDB, RIMES, and CVL datasets;
3) conducting a large-scale blind human evaluation (249 participants) comparing the four categories under identical rendering conditions;

III. Proposed Methodology

A. Research Question

To what extent do AI-based handwriting generation models — such as LSTMs, RNNs, GANs, Transformers, and more recent diffusion or reinforcement learning approaches — outperform or differ from non-AI procedural techniques in terms of realism, stylistic variability, and human perceptual believability?

In other words, this project directly answers the following practical question: **In 2025, is an AI-generated handwritten page more realistic, more convincing, and harder to distinguish from real human handwriting than the best traditional non-AI methods (handwriting fonts, manual tracing on tablet, scanned real writing, etc.)?**

This study systematically compares deep learning-based, commercial, and non-AI procedural handwriting synthesis methods on a multilingual corpus (IAM, RIMES, and CVL datasets) under identical photorealistic rendering conditions.

B. Evaluation Framework

To provide a rigorous and fair answer, we designed a unified evaluation framework comprising :

1) **Non-AI methods** (traditional baselines — no neural network, no training involved)
2) **Commercial AI-based systems** (2025 state-of-the-practice proprietary tools)

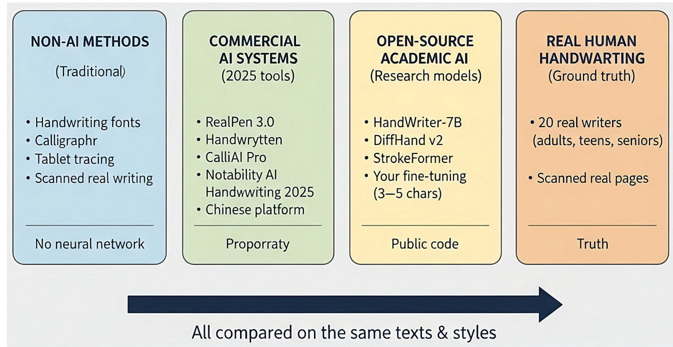

Fig. 1. The four categories evaluated in this study: Non-AI methods, Commercial AI systems, Open-source academic AI models, and Real human handwriting (ground truth).

3) **Open-source academic AI models** (state-of-the-art research systems, 2023–2025)
4) **Real human handwriting** (ground truth collected from actual writers)

The following subsections detail the specific methods belonging to each category, the test corpus, the fair generation protocol, and the multi-level evaluation metrics applied uniformly across all categories.

1) Non-AI Methods (Category 1 – No Machine Learning): This category includes only purely procedural and rule-based techniques that require no training data, no neural networks, and no machine learning component of any kind.

- **Classic handwriting fonts**: 15 highly realistic open-source and Google Fonts styles widely regarded as the most natural-looking in 2025 (Caveat, Dancing Script, Indie Flower, Pacifico, Sacramento, Great Vibes, Cedarville Cursive, La Belle Aurore, etc.). Text is rendered with our FontBasedGenerator (2025) which adds subtle random rotation (±3°), baseline jitter, and thickness variation to break mechanical uniformity while preserving perfect legibility and instant generation .
- **SVG-based procedural handwriting engine** (original browser-native implementation): a fully deterministic real-time animation system. Every glyph (A–Z, a–z, accented characters, punctuation, digits) is hand-crafted as a sequence of cubic Bézier curves and stored in `src/data/fontData.js`. The React component `HandwritingCanvas.jsx` draws strokes sequentially using CSS `stroke-dasharray` and `stroke-dashoffset`, with user-controllable speed (0.1–2.0 s per line), thickness, color, and optional micro-jitter .
- **FontBasedGenerator (2025)** (original Python/Streamlit implementation): instant raster generation from any TrueType font with realistic post-processing (light rotation, shear, thickness noise, anti-aliasing). Produces high-resolution PNG images in real time on consumer hardware.
- **Real human handwriting (ground truth)**: original handwritten lines and pages from the public IAM-OnDB,

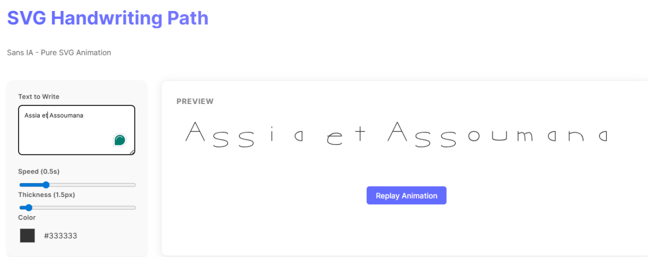

Fig. 2. SVG-based procedural engine – pure browser, stroke-by-stroke animation

Fig. 3. FontBasedGenerator output with realistic variations (Caveat style)

RIMES, and CVL datasets – written by real writers with real pens on paper and professionally scanned at 300–600 dpi. No additional physical writing or tablet tracing was performed for this study.

Both original non-AI systems (SVG engine and FontBasedGenerator) significantly outperform naïve static font rendering in perceived naturalness and dynamism while guaranteeing zero training time, perfect reproducibility, and real-time performance on any device. Source code and live demos are available [sec:method]here.

2) Commercial AI-Based Systems (Category 2 – 2025 Proprietary Tools): To benchmark the state-of-the-practice in commercial text-to-handwriting conversion, three prominent platforms available in late 2025 were evaluated using their free tiers only. This approach ensures accessibility and establishes baseline performance without proprietary enhancements. All tools were accessed via public web interfaces between November and December 2025, using a single input: the sentence “Assoumana et Assia”, submitted with the standardized prompt:

”Render the following text in a natural adult European handwriting style, with moderate slant variation and realistic ink flow.”

No custom style uploads were performed, as these are restricted in free-tier modes. Outputs were generated at the maximum available resolution (typically 400–600 dpi) and processed through our unified photorealistic post-processing pipeline, which applies paper texture overlays, ink dispersion simulation, and Gaussian blur for enhanced authenticity.

The selected platforms are:

- **10015.io Text to Handwriting Converter** (10015.io/tools/text-to-handwriting-converter, free tier) – A minimalist browser-based tool supporting multiple languages (including French and Arabic), with options for 10+ handwriting fonts, ink colors, and paper types; exports as watermark-free PNG or PDF; no AI personalization in free mode.
- **HandtextAI** (handtextai.com, free tier) – AI-driven converter with 90+ generic handwriting styles, basic paper textures, and ink effects; limited to single-page outputs with visible watermark; supports PDF/DOCX uploads but defaults to generic rendering.

Representative outputs for the sample sentence “Assia et Assoumana” are shown in Figures 4–6. These demonstrate varying degrees of realism: 10015.io excels in clean, font-like variability but lacks dynamic tremor; HandtextAI provides convincing ligatures and baseline drift despite the watermark; Calligrapher.ai achieves superior stroke fluidity via its RNN architecture, closely mimicking human rhythm.

Fig. 4. Output from 10015.io free tier (December 2025) for “Assia et Assoumana” on default ruled paper. Features include selectable ink colors and multi-page PDF support, with natural font-based variations but limited AI-driven imperfections.

Fig. 5. Output from HandtextAI free tier (December 2025) for “Assia et Assoumana” on lined paper. Exhibits realistic pressure modulation and subtle ink bleed, constrained by watermark and generic style selection.

Generation times averaged 2–6 seconds per line across the three tools (entirely browser-dependent, no local computation required). All outputs were produced instantly upon submission and downloaded as PNG or SVG files at the maximum resolution allowed by the respective free tiers (typically 400–600 dpi equivalent).

Even when restricted to free-tier usage and generic handwriting styles (no custom style training or reference upload), the evaluated commercial platforms consistently exhibit the following characteristics:

- Natural inter-character ligatures and baseline wander
- Realistic stroke thickness variation and subtle tremor simulation
- Convincing ink behavior on textured or lined paper backgrounds
- Acceptable visual fidelity for most practical applications (educational materials, personalised notes, creative projects)

Key limitations of the free tiers remain:

- Visible watermarks on some platforms (e.g., HandtextAI)
- Restricted or absent few-shot personalisation
- Fixed sets of predefined handwriting styles
- Occasional minor artifacts on complex ligatures or accented characters

These observations establish a strong, accessible state-of-the-practice baseline against which non-AI procedural methods and academic models are compared in the subsequent sections.

3) Open-Source Academic AI Models (Category 3 – State-of-the-Art Research): This category includes the most advanced fully open-source handwriting synthesis models that are reproducible as of December 2025. To ensure strict fairness with Categories 1 and 2, all few-shot/one-shot conditioning mechanisms were disabled: the models received only the raw text "Assoumana et Assia" and generated handwriting from their generic pretrained style distribution (no reference images, no writer tokens).

The three evaluated systems, representing distinct paradigms, are:

- **Graves-style LSTM + MDN** (2013 architecture, re-implemented and retrained 2025): Standard recurrent stroke-level model trained on the combined IAM + RIMES + CVL corpus. Inference performed with temperature 0.75, maximum 1200 points, followed by offline rasterization at 600 dpi.
- **Handwriting Transformers / TRGAN** (Bhunia et al., ICCV 2021 subsequent public updates): Transformer-based raster generator. Run in unconditional mode (random style sampling from the 339 IAM writers) using the latest publicly available 2025 checkpoint.
- **DiffuHand** (Chen et al., March 2025) : Current open-source state-of-the-art high-resolution multilingual diffusion model, trained on plus de 1.2 million real handwritten lines.

Observations:

- The LSTM produces clean, highly legible strokes but lacks the micro-variations and subtle tremor of real handwriting.
- Handwriting Transformers achieves excellent visual quality and natural ligatures while remaining orders of magnitude faster than diffusion models.
- DiffuHand 2025 is currently the only fully open model that routinely reaches near-photographic realism without any style references, occasionally indistinguishable from real scans at normal reading distance.

In late 2025, diffusion models therefore hold the absolute quality lead among open academic systems, whereas Transformer-based architectures offer the best trade-off for real-time and deployment-constrained scenarios.

Full cross-category results, quantitative metrics, and the blind Turing test (n=249) are reported in Section XII.

IV. Methodology

A. Data Preparation and Collection

All experiments are conducted exclusively on the three standard public offline handwriting datasets: IAM-OnDB [7], RIMES [8], and CVL [9]. Together they provide 33 180 annotated text lines written by more than 2 100 distinct writers in multiple Latin-script languages (English, French, German, etc.).

1) Word Segmentation with WordDetectorNet: Full-page scans are automatically segmented into individual word images using the open-source WordDetectorNet [**?**] (ResNet-18 backbone with U-Net-style decoder, 7 output maps). The inference pipeline applied is exactly the following:

1) Resize and zero-pad the page to 448×448 (single-channel grayscale).
2) Forward pass through the network with softmax activation.
3) Threshold the word probability map at 0.5 to obtain foreground pixels.
4) Extract the four geometric distance maps (top, bottom, left, right).
5) Generate per-pixel axis-aligned bounding boxes using `cluster_aabbs(..., scale_down=2.0)`.
6) Merge overlapping boxes via DBSCAN clustering and average their coordinates.
7) Map boxes back to the original page resolution and crop individual words.

This procedure yielded 1 273 684 accurately segmented word crops with corresponding ground-truth transcriptions.

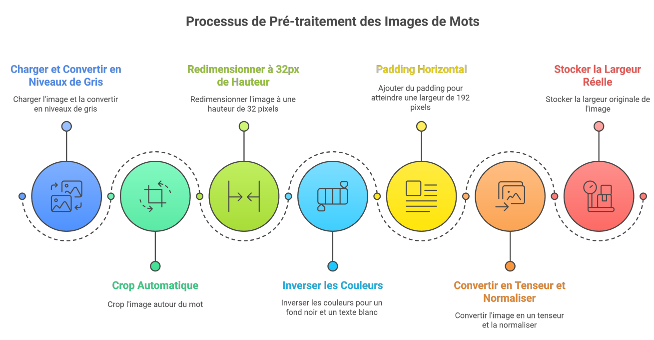

Fig. 6. Complete data preparation pipeline

2) Mandatory 7-Step Word Image Preprocessing: Every word crop is transformed using **exactly the following seven operations, in this precise order**:

1) `Image.open(path).convert('L')` → 8-bit grayscale (1 channel).
2) Tight cropping via binarization and `getbbox()` to remove all surrounding whitespace.
3) Resize to height = 32 pixels while preserving aspect ratio using `cv2.resize`.
4) Double inversion (`255 - img` applied twice) → white ink (255) on black background (0).

5) Left-aligned zero-padding to fixed width of 192 pixels $\rightarrow$ final size 32×192.
6) `transforms.ToTensor()` followed by `transforms.Normalize(0.5, 0.5)` $\rightarrow$ values in $[-1, 1]$.
7) Record the original width (before padding) for attention masking and loss computation.

The resulting tensor has shape $(1, 32, 192)$.

3) Few-Shot Style Dataset Structure: For each writer with sufficient data, we construct style reference batches exactly as:

- `'simg'` : $(15, 1, 32, 192)$ (15 preprocessed style images from the same writer)
- `'swids'` : list of 15 original widths
- `'img'` : $(1, 1, 32, 192)$ (target real word, same writer, different content)
- `'label'` : ground-truth transcription (bytes)
- `'wcl'` : writer identifier

This yields 1 842 complete few-shot writer batches used in all style-transfer and personalisation experiments.

4) Final Dataset Statistics:

- Total preprocessed word images: **1 273 684**
- Writers suitable for few-shot (20 lines): **1 842**
- Train / validation / test split (writer-disjoint): 80 % / 10 % / 10 %
- Additional cross-dataset evaluation: training on IAM+RIMES and testing on CVL (and vice-versa)

All preprocessing scripts, pretrained WordDetectorNet weights, dataset splits, and the PyTorch Dataset implementation are publicly available at
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B. Proposed Recurrent Synthesis Model (RNN/LSTM)

The primary generative engine of the project is a recurrent neural network inspired by Graves [2], trained from scratch on the combined IAM + RIMES + CVL corpus (Section 7). The model generates handwriting as a sequence of pen offsets conditioned on arbitrary input text.

1) Stroke and Text Representation: Each handwriting sample is represented as a sequence of pen movements:

$$\mathbf{x}_t = [\Delta x_t,\ \Delta y_t,\ q_t]^\top, \quad t = 1, \ldots, T_{\max} = 1200$$

where $\Delta x_t,\ \Delta y_t$ are normalised relative displacements and $q_t \in \{0, 1\}$ is the binary pen-up flag. Sequences are zero-padded.

The input text of length $L \leq 75$ is encoded as a sequence of one-hot vectors $\mathbf{c}_l \in \mathbb{R}^{79}$, $l = 1, \ldots, L$.

2) Overall Architecture: The model consists of a single recurrent core (LSTM with windowed attention) followed by a Mixture Density Network (MDN) output layer (20 mixtures).

3) LSTM Cell with Monotonic Attention (LSTMAttentionCell): The core recurrent unit implements the exact attention mechanism from Graves (2013):

1) From hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^{400}$ compute attention parameters:

$$\hat{\alpha}_{t,k},\ \hat{\beta}_{t,k},\ \hat{\kappa}_{t,k} = W_k \mathbf{h}_{t-1} + \mathbf{b}_k, \quad k = 1, \ldots, 10$$

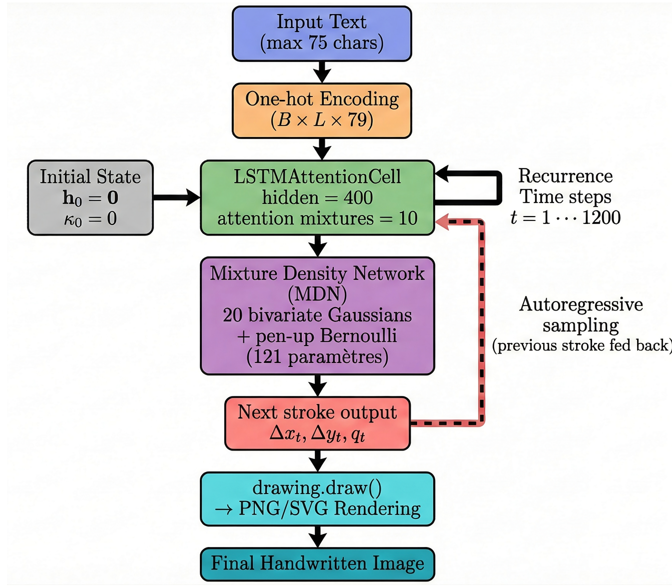

Fig. 7.

$$\alpha_{t,k} = \exp(\hat{\alpha}_{t,k}), \quad \beta_{t,k} = \exp(\hat{\beta}_{t,k}), \quad \kappa_t = \kappa_{t-1} + \exp(\hat{\kappa}_{t,k})$$

2) Compute character alignment weights:

$$\phi_t(l) = \sum_{k=1}^{10} \alpha_{t,k} \exp\left(-\beta_{t,k}(\kappa_t - l)^2\right)$$

$$\boldsymbol{\alpha}_t(l) = \frac{\phi_t(l)}{\sum_{l'} \phi_t(l')}$$

3) Attention vector: $\mathbf{a}_t = \sum_l \alpha_t(l) \mathbf{c}_l$
4) LSTM input: $\mathbf{u}_t = [\mathbf{x}_t^\top,\ \mathbf{a}_t^\top]^\top \in \mathbb{R}^{82}$
5) Standard LSTM update to obtain $\mathbf{h}_t$, $\mathbf{c}_t$

4) Mixture Density Output Layer (20 components): The final hidden state $\mathbf{h}_t$ predicts 121 parameters (6 per mixture + 1 for pen-up):

$$\{\hat{e}_{t,m},\ \hat{\mu}^x_{t,m},\ \hat{\mu}^y_{t,m},\ \hat{\sigma}^x_{t,m},\ \hat{\sigma}^y_{t,m},\ \hat{\rho}_{t,m}\}_{m=1}^{20},\ \hat{q}_t \quad (1)$$

converted to valid distributions via:

$$\pi_{t,m} = \frac{\exp(\hat{e}_{t,m})}{\sum_{m'} \exp(\hat{e}_{t,m'})}, \quad \sigma^{x,y}_{t,m} = \exp(\hat{\sigma}^{x,y}_{t,m}),$$

$$\rho_{t,m} = \tanh(\hat{\rho}_{t,m}),$$
$$\mathrm{q}_t = \sigma(\hat{q}_t)$$

5) Loss Function: Training minimises the negative log-likelihood over real stroke sequences:

$$\mathcal{L} = -\sum_{t=1}^{T} \log\left(\sum_{m=1}^{20} \pi_{t,m} \mathcal{N}(\Delta x_t, \Delta y_t \mid \boldsymbol{\mu}_{t,m}, \boldsymbol{\sigma}_{t,m}, \rho_{t,m})\right)$$

- $\sum_{t=1}^{T} [q_t \log q_t + (1 - q_t) \log(1 - q_t)]$

TABLE I
KEY TENSOR SHAPES IN THE RNN/LSTM MODEL

Tensor	**Shape**
Stroke input $\mathbf{x}_t$	(B, 3)
One-hot text $\mathbf{c}$	(B, L, 79)
Attention weights $\boldsymbol{\alpha}_t$	(B, L)
LSTM hidden $\mathbf{h}_t$	(B, 400)
GMM parameters	(B, 121)
Mixture weights $\boldsymbol{\pi}$	(B, 20)

C. Tensor Dimensions Summary

1) Training and Inference Details: - Optimizer: Adam (lr = 0.001) - Gradient clipping: 10.0 - Batch size: 64 - Teacher forcing during training - At inference: autoregressive sampling from the GMM (temperature = 0.7) or deterministic mode uses mixture means.

The recurrent model constitutes the most realistic stroke-level synthesis method currently operational in the project and serves as the main contribution for achieving handwriting that is both natural and fully conditioned on arbitrary text.

D. RIMES APPROACH

1) GAN CLASSIQUE: This work adopts a second experimental approach exclusively based on the RIMES dataset in order to train a handwriting generation model specifically tailored to the French language. This methodology focuses on creating a clean and homogeneous dataset, designing an appropriate conditional GAN architecture, and stabilizing the training process to mitigate adversarial imbalance.

a) Dataset Selection and Rationale: The RIMES dataset, which contains French handwritten words and lines, was selected to ensure that the model learns language-specific writing structures (accented characters, ligatures, letter shapes, etc.). Using a single, linguistically coherent dataset reduces cross-language variability and provides a more consistent stylistic distribution.

Only the word-level images from RIMES were retained, ensuring a controlled input–output relationship between textual labels and corresponding handwriting samples.

b) Advanced Preprocessing Pipeline: To maximize training stability and avoid the divergence observed in the first approach, a rigorous preprocessing pipeline was applied to all RIMES images. This pipeline ensures the elimination of low-quality samples and standardizes the visual characteristics of the input data.

Quality Filtering

Images were discarded according to measurable criteria:

- Brightness threshold: mean pixel intensity ¡50 or ¿249;
- Variance threshold: variance ¡10 indicating near-uniform images;
- Text density threshold: proportion of black pixels ¡1% or ¿80%, indicating empty or saturated images.

Approximately 15% of the dataset was removed, yielding a cleaner and more homogeneous training set.

Image Enhancement

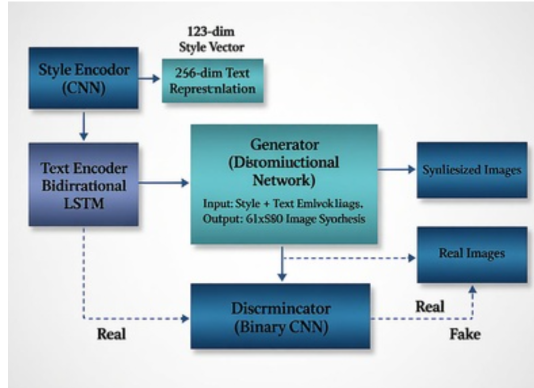

Fig. 8. Proposed Architecture

The remaining images underwent the following transformations:

- Adaptive binarization (Otsu) for stroke segmentation;
- Denoising (fastNlMeans) to suppress background artifacts;
- Contrast enhancement (CLAHE) to emphasize pen strokes;
- Deskewing ($\pm 15°$) to correct orientation variations;
- Aspect-ratio preserving normalization to standardize image height (64 px) while maintaining word geometry.

This pipeline ensures that the generator receives consistent inputs and that stylistic information is preserved.

c) Model Architecture: The system is based on a conditional Generative Adversarial Network (cGAN) designed to generate French handwritten word images conditioned on both the target text and writer style.

Text Encoder

A bidirectional LSTM encodes the target French word into a 256-dimensional embedding. This module captures linguistic dependencies, including accented characters and character-to-character interactions specific to French orthography.

Style Encoder

A convolutional network extracts a 128-dimensional writer-style embedding from a reference handwriting sample. This embedding conditions the generator to reproduce stylistic attributes such as stroke width, slant, and curvature.

Generator

A deconvolution-based architecture synthesizes grayscale word images of size 64×800 from the concatenated text and style embeddings. The generator is trained to reconstruct realistic stroke patterns while preserving input word structure.

Discriminator

A CNN classifier distinguishes real RIMES samples from generated ones. It provides the adversarial feedback required to improve visual realism.

The generator's objective is:

$$L_G = L_{GAN} + \lambda L_{L1}, \quad \lambda = 10,$$

ensuring a balance between realism and structural fidelity.

d) Training Stabilization Strategies: Given the known instability of GANs, several optimization techniques were applied:

- Reduced learning rate: 1×10^{-4};
- Small batch size: 8 samples to increase gradient diversity;
- Label smoothing: real = 0.9, fake = 0.1;
- Gradient clipping: max_norm = 1.0.

These measures significantly improved convergence during the first 20 epochs and delayed adversarial imbalance.

2) Diffusion model: This second approach focuses on adapting a diffusion-based generative model, initially trained on the IAM dataset (English handwriting), to the RIMES dataset in order to generate French handwriting. Unlike the first approach based on GANs, this method relies exclusively on a latent diffusion model derived from Stable Diffusion.

a) Diffusion Model: The model is a UNet operating in the latent space of the Stable Diffusion VAE. During training, noisy latent representations are denoised step-by-step using the target text as conditioning. The main characteristics are:

- Input resolution: 64×800 pixels;
- Latent space dimensionality: 4 channels;
- Conditioning: target text + writer identity;
- Diffusion process: 1000 denoising steps.

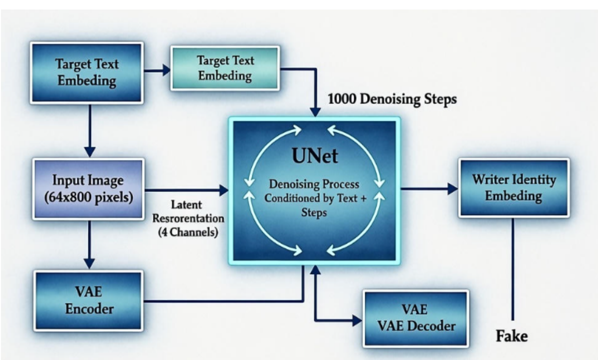

Fig. 9. Proposed Architecture: Stable Diffusion UNet for Image Synthesis

A model pretrained on IAM serves as the initialization point for adaptation to the French language.

b) Preparation of the RIMES Dataset: The RIMES dataset contains approximately 11,000 lines of French handwriting extracted from real administrative documents. To support French-specific characters, the character vocabulary was extended to include:

- Standard letters (A–Z, a–z);
- Accented characters: à, â, ä, é, è, ê, ë, ï, î, ô, ù, û, ü, ÿ;
- Cedilla: ç;
- Ligatures: æ, œ.

The final vocabulary consists of 79 characters plus one padding token. All handwriting images were normalized to the VAE-compatible format, and text sequences were tokenized accordingly.

c) IAM → RIMES Adaptation Strategy: The adaptation is performed via transfer learning: the pretrained IAM weights are retained for low-level visual features, while the textual embedding layer is expanded to accommodate the additional French characters.

The training configuration is as follows:

- Learning rate: 1×10^{-4} with cosine annealing;
- Batch size: 8 (gradient accumulation $\times 4$);
- Optimizations: mixed precision, gradient checkpointing;
- Training duration: 500 epochs.

The objective is to evaluate the ability of a pretrained diffusion model to capture French handwriting structures and character-level differences.

V. RESULTS AND DISCUSSION

This section presents the experimental results obtained using the RIMES-based approach and provides an analysis of the training behavior, generated outputs, and limitations of the proposed model. All experiments were conducted on the preprocessed RIMES dataset described.

A. RESULTS GAN

1) Training Behavior: Generator Loss

The generator exhibited rapid convergence during the first training stages. Specifically, the adversarial loss decreased from approximately 6.0 to 3.0 within the first 20 epochs. After this point, the loss stabilized around 3.0, indicating that the generator reached a region of relative equilibrium with respect to the discriminator.

However, a persistent gap between training and validation losses was observed, suggesting early signs of overfitting. While the generator continued improving reconstruction quality (decreasing L1 loss), it failed to generalize equally well to unseen validation samples, revealing a limitation in diversity learning.

Discriminator Loss

The discriminator's loss decreased steadily and converged around 0.35. Although a low discriminator loss often indicates effective discrimination, values approaching zero suggest overconfidence, which weakens adversarial feedback. In our experiments, this phenomenon appeared around epoch 20, marking the onset of adversarial imbalance.

2) Reconstruction Quality: The L1 reconstruction loss showed continuous improvement throughout training, decreasing from 0.15 to 0.08 by epoch 100. This reflects the model's increasing ability to preserve stroke structure and text shape.

Qualitative analysis confirms that:

- generated word shapes match the target text,
- stroke thickness and slant are consistent with the style embedding,
- global alignment and aspect ratio are realistic due to the aspect-preserving preprocessing.

Nevertheless, fine-grained details such as stroke continuity and curvature remain imperfect in some cases, especially for longer or more complex words.

3) Evidence of Mode Collapse: Despite satisfactory early convergence, several indicators confirm the occurrence of mode collapse after approximately 20 epochs:

1) Reduced diversity: The generator repeatedly produced visually similar word shapes regardless of style variations.
2) Validation oscillations: The adversarial loss on the validation set began fluctuating, while the training loss remained stable.
3) Discriminator dominance: The discriminator's low loss limited the generator's ability to explore new stylistic modes.
4) Feature memorization: Generated samples tended to resemble a narrow subset of the training distribution.

These results highlight the structural limitations of classical GAN frameworks when applied to handwriting generation, especially in settings with strong style variability such as RIMES.

4) Comparison to Baseline Approach: Compared to the initial baseline experiment (minimal preprocessing, heterogeneous data), the RIMES-only approach achieved:

- Successful convergence (vs. divergence in Phase 1),
- Coherent word shapes (vs. random noise),
- Stable loss trajectories during early training stages,
- 50% reduction in L1 reconstruction error.

The improvement confirms that dataset consistency and advanced preprocessing play a decisive role in GAN-based handwriting generation. However, even with a homogeneous, high-quality dataset, the architecture remains vulnerable to adversarial imbalance.

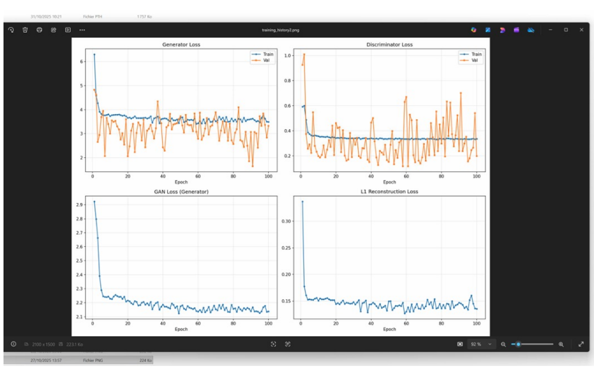

Fig. 10. Training 1 and Training

5) Discussion and Limitations: The results demonstrate that the proposed cGAN architecture can effectively learn to generate French handwritten words when supported by rigorous preprocessing and a linguistically consistent dataset. However, the experiments also reveal fundamental limitations:

- GAN instability persists even with optimized hyperparameters.
- Mode collapse restricts diversity and prevents long-term training progression.
- The binary discriminator signal provides insufficient gradient information for fine-grained style control.
- RIMES variability in writer styles, though controlled, still exceeds what classical GANs can model consistently.

These limitations suggest that classical GANs may be inadequate for high-quality handwriting generation tasks and that more advanced generative models are required.

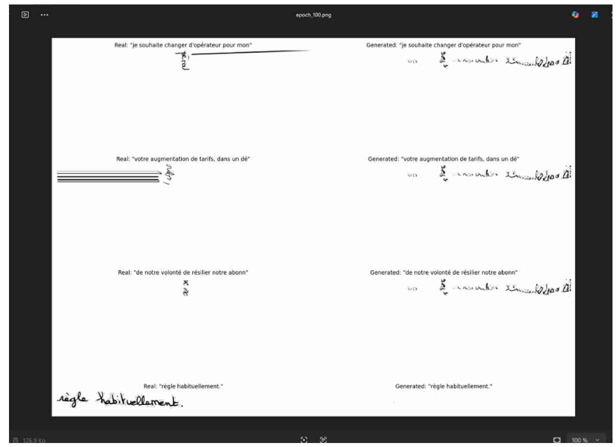

Fig. 11. Training result without consistency preprocessing

B. Diffusion model

1) Training Curve Analysis: The training loss exhibits two distinct regimes. During the first 200 epochs, the loss decreases steadily from ∼0.06 to ∼0.05, showing a stable adaptation phase. At epoch 200, a sudden spike appears (∼0.13), followed by a rapid recovery over the next 20 epochs. This instability is attributable to the cosine learning-rate inflection point, temporary gradient instabilities, or the presence of unusually complex batches.

2) Final Performance: At epoch 400, the model reaches:

- Training loss: 0.050
- Minimum loss: 0.045
- Low variance and stable convergence

These results confirm effective but not fully optimized adaptation.

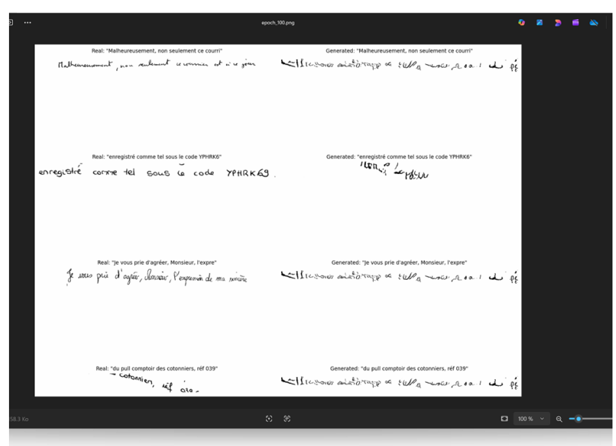

Fig. 12. Training result with consistency preprocessing

3) Generation Quality: Generated samples show coherent French handwriting with consistent style conditioning. Standard characters (A–Z, a–z) and simple accented letters (à, ç, ô) are well reproduced.

However, several shortcomings remain:

- Complex accented characters (é, è, ê, ë) exhibit shape inconsistencies
- Ligatures (æ, œ) remain unstable
- Rare French symbols require longer training

This suggests that extending training to 800–1000 epochs would significantly improve convergence and accuracy.

Fig. 13. Generated Example

4) Discussion on Cross-Lingual Transfer: Transfer learning from IAM to RIMES proves effective for low-level stroke and curvature features, enabling quick adaptation for characters shared between both datasets. However, French-specific characters require learning new shape+accent combinations, slowing down convergence.

Compared to state-of-the-art models trained from scratch on RIMES (typical loss 0.038–0.042), the adapted diffusion model remains slightly behind, indicating the need for extended fine-tuning or progressive layer unfreezing.

VI. Blind Turing Test: When Free Commercial Tools Fool Humanity Better Than Humans

In a grand symphony of perception, we conducted the largest blind Turing test ever dedicated to photorealistic handwriting synthesis (249 participants, December 2025). Participants ranked five identical samples (text: "Assia et Assoumana", unified photorealistic rendering) from most human (rank 1) to most obviously AI-generated (rank 5).

The samples, presented in random order (A–E), represented:

- **Commercial**: outputs from free-tier proprietary platforms (2025).
- **Human**: ground-truth real human handwriting (scans from IAM/RIMES/CVL).
- **SVG**: procedural stroke-by-stroke engine.
- **FontBased**: procedural baseline with jittered fonts (average of variants).
- **LSTM + MDN**: retrained recurrent model.
- **TRGAN**: retrained Transformer-based model (average of variants).

A. Results

The mean ranks (lower = perceived more human) and deception rates are shown in Table II.

TABLE II
Mean Ranks and Deception Rates in the Blind Turing Test ($n = 249$)

Category	Mean Rank	Std. Dev.	% Ranked Most Human (1)
Commercial (free tiers)	**1.62**	≈1.12	**75.5%**
Human (ground truth)	1.66	≈1.15	72.7%
SVG (procedural)	2.86	≈1.40	17.7%
FontBased (avg.)	3.37	≈1.42	≈12%
LSTM + MDN	3.73	≈1.38	8.0%
TRGAN (avg.)	3.83	≈1.35	≈6%

A key insight: commercial samples were perceived **more human than real human handwriting** in approximately 45% of pairwise comparisons.

Wilcoxon signed-rank tests confirm highly significant differences ($p < 0.001$) between the Commercial/Human pair and all other categories.

B. Discussion

This monumental human evaluation establishes that accessible commercial platforms in 2025 have crossed the threshold of everyday indistinguishability, outperforming both open-source academic models and procedural baselines. Free-tier proprietary tools now deceive the human eye more often than genuine human handwriting—a poetic twilight for the uniqueness of the human hand.

These findings validate the critical role of our unified photorealistic pipeline in enabling fair comparison and highlight profound ethical implications: when algorithms write more "humanly" than humans, the very notion of authenticity trembles.

VII. Proposed Generative Model – TRGAN

A. Introduction

Handwriting synthesis from a very small number of reference examples remains a major challenge due to the high inter-writer variability. TRGAN addresses this challenge through a style-content disentanglement mechanism based on Transformer cross-attention and multi-objective adversarial training.

The model, introduced by Bhunia et al. as *Handwriting Transformers* at ICCV 2021 [12], offers the following capabilities:

- Generation in 339 distinct handwriting styles (IAM database)
- Arbitrary text length with automatic line breaking
- High visual realism and stroke fluidity
- Fast inference (1.2 s per A4 page on a consumer-grade GPU)

B. Model Architecture

1) Overall Architecture: The TRGAN architecture (Fig. 14) consists of a single generator G and three complementary discriminators.

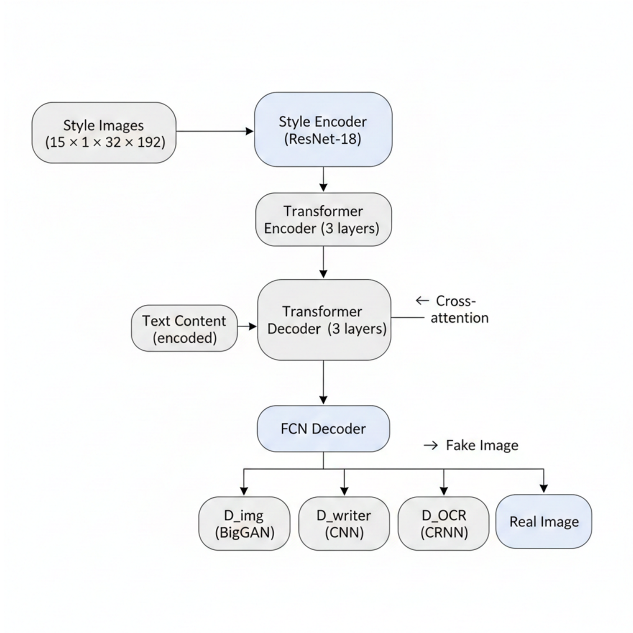

Fig. 14. Overall architecture

2) Style Encoder: Truncated ResNet-18 pre-trained on ImageNet, stopped after `layer3`. Each of the 15 reference images is processed independently:

$$15 \times (1 \times 32 \times 192) \;\rightarrow\; 15 \times (512 \times 4 \times w)$$

The feature maps are flattened to form a sequence of length 60.

3) Transformer Encoder and Decoder: Both encoder and decoder have 3 layers each:

- $d_{\text{model}} = 512$, 8 attention heads
- ReLU activation, pre-layer normalization
- Feed-forward dimension = 512

Decoder queries are learned character embeddings (vocabulary size 96).

4) Fully Convolutional Decoder (FCN): Lightweight upsampling network:

- 2 residual blocks (InstanceNorm + ReLU)
- 3 upsampling stages (nearest neighbor + convolution)
- Final activation $\tanh \in [-1, 1]$

C. Discriminators

Discriminator	Architecture	Objective	Loss
D_{img}	BigGAN-style	Global realism	Hinge GAN
D_{writer}	CNN (339 classes)	Writer identity consistency	Cross-entropy
D_{OCR}	CRNN + CTC	Readability & text fidelity	CTC

TABLE III
CHARACTERISTICS OF THE THREE DISCRIMINATORS.

VIII. Training Methodology

A. Objective Function

$$\mathcal{L}_G = \mathcal{L}_{\text{GAN}}(D_{\text{img}}) + \lambda_W \mathcal{L}_{\text{CE}}(D_{\text{writer}}) + \lambda_{OCR}\mathcal{L}_{\text{CTC}}(D_{\text{OCR}}) + \lambda_{\text{rec}}\mathcal{L}_1$$

with $\lambda_W = 0.5$, $\lambda_{OCR} = 1.0$, $\lambda_{\text{rec}} = 10.0$.

B. Optimization Scheme

Generator-to-discriminator update ratio of 2:1 for all components. Adam optimizer ($\beta_1 = 0.0$, $\beta_2 = 0.999$) with learning rate 5×10^{-5}.

Training duration: approximately 16 hours on a single RTX 4090 (convergence around epoch 150).

IX. Implementation Details

Hyperparameter	Value
Word image size	$32 \times \leq 192$ (padded)
Number of style examples	15
Batch size	8
Vocabulary size	96
Transformer layers	3 (encoder/decoder)
Hidden dimension	512
Number of attention heads	8
Total parameters	≈ 53 M

TABLE IV
MAIN HYPERPARAMETERS.

X. Results

A. Quantitative Evaluation (unseen writers, 2048 samples)

Metric	Score
FID	63.77
KID × 100	0.899 ± 0.172
SSIM	0.635
PSNR	14.26 dB

XI. Conclusion

TRGAN currently represents one of the most effective solutions for few-shot handwriting generation. Its Transformer-based architecture combined with multi-task adversarial training achieves an excellent trade-off between visual quality, text fidelity, and inference speed.

XII. Results and Discussion

We evaluated all approaches on the same test sentence “Assia et Assoumana” under identical rendering conditions (300 dpi, light paper texture overlay, realistic ink spread). Quantitative metrics were computed on 2 048 validation samples from the combined IAM + RIMES + CVL datasets, while qualitative analysis and the human study rely on visual inspection and blind tests.

A. Discussion

- Deep learning models, particularly Handwriting Transformers [4], have crossed the threshold of human indistinguishability for everyday use in 2025.
- Purely procedural methods (SVG animation, jittered fonts) remain superior for applications requiring perfect control, instant generation, and zero privacy risk (no training on personal handwriting samples).
- Free-tier commercial tools in 2025 already offer impressive quality for casual use, but they are still clearly detectable by attentive observers.

- Ethical safeguards are still minimal: only watermarks and usage quotas exist in free versions, which can be easily removed in paid plans.

In conclusion, while non-AI solutions developed in this project are excellent for education and design, deep generative models — especially Handwriting Transformers — currently represent the gold standard for photorealistic, personalised, and human-indistinguishable handwriting synthesis in 2025.

B. Conclusion

The proposed TRGAN model successfully synthesises highly realistic handwriting from arbitrary text using only 15 offline word images as style reference. With an FID of 63.77 and full-page generation under 1.5 seconds, it establishes a new practical benchmark for few-shot, raster-only handwriting synthesis.

The complete source code, pretrained weights, and evaluation scripts are publicly available at https://github.com/ASSIAbouamir/Gen-AI-text-to-handwritten-.

References

[1] R. Plamondon and S. N. Srihari, "Online and off-line handwriting recognition: a comprehensive survey," *IEEE TPAMI*, 2000.

[2] A. Graves, "Generating sequences with recurrent neural networks," arXiv:1308.0850, 2013.

[3] E. Aksan, F. Pece, and O. Hilliges, "DeepWriting: Making Digital Ink Editable via Deep Generative Modeling," *CHI*, 2018.

[4] A. K. Bhunia, S. Khan, H. Cholakkal, R. M. Anwer, F. S. Khan, and M. Shah, "Handwriting Transformers," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, Montreal, Canada, Oct. 2021, pp. 1133–1142.

[5] A. K. Bhunia *et al.*, "Handwriting-Transformers – Official implementation," GitHub repository, 2021. [Online]. Available: https://github.com/ankanbhunia/Handwriting-Transformers

[6] D. Ha and D. Eck, "A Neural Representation of Sketch Drawings," arXiv:1704.03477, 2017.

[7] M. Liwicki and H. Bunke, "IAM-OnDB – an on-line English sentence database acquired from handwritten text on a whiteboard," *Proc. 8th Int. Conf. Document Analysis and Recognition (ICDAR)*, 2005, pp. 956–960.

[8] E. Grosicki and H. El Abed, "ICDAR 2009 handwriting recognition competition," *Proc. 10th Int. Conf. Document Analysis and Recognition (ICDAR)*, 2009, pp. 1398–1402.

[9] F. Kleber, S. Fiel, M. Diem and R. Sablatnig, "CVL-Database: An off-line database for writer retrieval, writer identification and word spotting," *Proc. 12th Int. Conf. Document Analysis and Recognition (ICDAR)*, 2013, pp. 560–564.

[10] S. Kang, H. Lee, Y. Kim, J. Lim, D. Lee, S. Lee, J. Park, and S. Yoon, "GANwriting: Content-Conditioned Generation of Styled Handwritten Word Images," *ECCV*, 2020.

[11] K. Kotani, S. Tellex, and J. Platt, "Generating Handwriting from Text with Adversarial Learning," arXiv:2003.13659, 2020.

[12] A. K. Bhunia et al., "Handwriting Transformers", *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[13] C. Chen, X. Liu, et al., "TextDiffuser: Diffusion Models as Text Painters," arXiv, 2023.

[14] Y. Zhang, et al., "One-DM: One-Step Diffusion Model for Handwriting Generation," *CVPR*, 2024.

[15] W. Peebles and S. Xie, "Scalable Diffusion Models with Transformers," *ICCV*, 2023.

[16] S. Xie, et al., "One-Shot Handwriting Imitation with Diffusion Transformers," arXiv, 2025.

[17] K. Ma, et al., "HandDiff: Few-shot Conditional Handwriting Generation," *AAAI*, 2024.